Avyya Singh

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Education

McMaster University Sept 2022 – Apr 2027
Bachelor's of Computer Science (Co-op) *Hamilton, Canada*

- Relevant Coursework: Data Structures & Algorithms, Software Engineering Principles, OOP & Software Development, AI/Machine Learning, Computer Architecture, Computer Networks and Security.
- **General Executive, McMaster Cyber Society** (Mar 2025 – Present): scaled the society to 100+ members, ran weekly hands-on CTF training (web exploitation, forensics, reverse engineering), and led the team to **3rd in Canada at CyberSci 2025** (2nd in the Toronto Region).

Experience

Associate Machine Learning Developer Intern May 2026 - Aug 2026
ServiceNow *Toronto, Canada*

- Designed and developed the first internal **MCP server** for an **AI-powered optimization platform**, architected to scale across a large production database and support customer datasets of **90,000+** records and beyond, reducing debugging and investigation time from days to minutes.
- Enhanced the platform's **ML optimization engine** by maintaining production ML pipelines, improving optimization logic, and implementing customer-driven features and configurable optimization parameters using **Python, Pydantic, Docker, and Git**.

Software Engineering Intern Jan 2026 - Apr 2026
AMD (Advanced Micro Devices) *Markham, Canada*

- Reduced manual benchmarking time by 15 hours/week by building automation scripts that cut individual validation runs from hours to under 5 minutes.
- Enhanced **ML infrastructure** and evaluation tooling, developing features for ML-driven game models and training workflows across **Roblox and CS:GO**, including movement, navigation, and combat behaviors, used for automated stress-testing and performance benchmarking.
- **Parallelized benchmarking workflows** across **100+ machines** in the **GPU validation lab**, reducing total validation time across large-scale hardware runs.
- Added new features to the internal ML benchmarking tooling, including automated detection of customer-facing errors prior to release.

Associate Machine Learning Developer Intern July 2025 - Dec 2025
ServiceNow *Toronto, Canada*

- Developed and maintained **ML pipelines** and tooling with runtime-configurable experiments and robust data generation; leveraged **Python, Docker, and Git** to deliver reliable, reproducible workflows.
- Elevated engineering quality with automated testing and **CI/CD (pytest, pre-commit, Jenkins)** and lightweight **Bash scripting** to streamline developer workflows and reduce release friction.

Software Engineering Intern May 2024 – Aug 2024
ECSite, Inc. *Remote, USA*

- Developed a **full-stack project**, transforming Excel spreadsheets into an interactive dashboard using **Node.js, Tailwind CSS, and Material UI** for enhanced data visualization.
- Tested and deployed the full-stack project in **E1** and **E2** environments, utilizing **CI/CD pipelines** with **GitHub Actions and Docker** to automate development workflows.

Projects

Cardio Disease Prediction in ML | *Python, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn*

- Designed and implemented predictive models, including **Logistic Regression, Decision Tree, KNN, and SVM**, to classify cardiovascular disease risk using demographic, clinical, and lifestyle data.
- Preprocessed a dataset of 70,000 samples by standardizing features. Achieved evaluation through **5-fold cross-validation** (accuracy, precision, recall, F1-score, and ROC-AUC).

Technical Skills

Languages: Python, JavaScript, Java, C/C++, SQL, HTML/CSS, R, Haskell, Bash/Shell
Frameworks: React.js, Next.js, TypeScript, Node.js, Angular, PyTorch, Flask, Pydantic, NumPy, Pandas, Click
Developer Tools: Git, Docker, Kubernetes, Azure, AWS, Power BI, Tableau, pytest, pre-commit, Jenkins